Conventional pansharpening

Launcher: `Toolbox/pansharpening.bat`

Purpose

Pansharpening transfers the finer spatial detail of a single-band panchromatic (PAN) image to a lower-resolution multispectral (MUL) image. It can make small crop-mark patterns easier to see, but it creates a derived image and can introduce spectral or edge artefacts.

Inputs and parameters

Option	What to provide	Behaviour
Input MUL Image	Corrected multiband GeoTIFF	The image is aligned to the PAN grid internally. Use the matching acquisition and footprint.
Input PAN Image	Corrected single-band panchromatic GeoTIFF	Provides the output grid and spatial detail. It should share the MUL image's CRS and acquisition.
Output Folder	Folder, or blank	Blank writes beside the MUL image. A missing entered folder is created only at its final level.
Vector layer	Optional clip polygon	Crops the final pansharpened result and assigns NoData `-9999`. The unclipped temporary result is removed.
Sensor	`WV3` or `LEGION`	Used by weighted Brovey. Default: `WV3`. Sensor choice does not affect Bayesian processing.
Algorithm	`Bayesian` or `wBrovey`	Selects the fusion method. Default: `Bayesian`.

Option	What to provide	Behaviour
lambda	Floating-point value from 0 to 1	Bayesian regularization/weighting parameter. Default: `0.995`. The field is validated even when `wBrovey` is selected.

Algorithms

Bayesian uses Orfeo ToolBox's Bayesian method. The MUL raster is aligned with nearest-neighbour resampling. Higher lambda values impose a stronger PAN-driven constraint. The default is intended for reflectance scaled to integer-like values; with unscaled 0–1 reflectance, it can overwhelm the spectral signal and produce nearly identical bands or extreme values.

wBrovey injects PAN detail using sensor-specific spectral weights and linear MUL resampling. WV3 uses seven weighted bands, then pansharpens NIR2 separately with local mean/variance matching and creates a VRT stack. The current LEGION sensor has no Brovey weights configured, so `wBrovey` is not operational for LEGION; use Bayesian instead.

Outputs

- Bayesian: `<MUL-name>_bayes.tif`
- Weighted Brovey: `<MUL-name>_brovey.tif`
- WV3 weighted Brovey also creates an NIR2 LMVM product and a `HIGHRES_stack.vrt` that combines the products.

Main outputs are DEFLATE-compressed Cloud Optimized GeoTIFFs. Intermediate alignment and calculation rasters are deleted after successful processing.

Recommended settings and checks

- For DOS3 outputs scaled by `10000`, begin with Bayesian lambda `0.995`.
- For reflectance in the 0-1 range, begin near `0.1`, then compare results rather than assuming one value is universally correct.
- Check that output bands have plausible and distinct min/max, mean, and standard deviation values. Nearly identical statistics in every band indicate spectral collapse.

- Compare sharp linear features with the original PAN. Doubled edges, halos, and shifts may indicate input misregistration rather than archaeological structure.
- Keep the unsharpened multispectral data for quantitative spectral analysis; pansharpened pixels are modelled values.